

Chapter 1. Belt & Rope Drives

1. Slip & Creep of belt
2. Equation on Length of Open belt drive
3. Equation on Ratio of Driving Tensions for flat belt drive
4. Equation on Centrifugal Tension
5. Equation on Initial Tension
6. Condition for Transmission of maximum power
7. Material used for belt

Chapter 2. Toothed Gearing

1. Gear terminology
2. Difference between Cycloidal teeth & involute teeth
3. Equation on Length of Path of Contact
4. Interference in Involute Gears
5. Minimum Number of Teeth on the Pinion in Order to Avoid Interference
6. Centre Distance for a Pair of Spiral Gears
7. Efficiency of Spiral Gears

Chapter 3. Gear Train

1. Comparison between different types of gear train
2. Differential gear box
3. Epicyclic gear train
4. Compound Epicyclic Gear Train—Sun and Planet Gear

Chapter 4. A. Gyroscope Principle of Gyroscopic Couple

1. Effect of the Gyroscopic Couple on an Aeroplane
2. Effect of Gyroscopic Couple on a Naval Ship during Steering, Pitching & Rolling

Chapter 4. B. Flywheel

1. Determination of Maximum Fluctuation of Energy
2. Coefficient of Fluctuation of Energy
3. Coefficient of Fluctuation of Speed
4. Energy Stored in a Flywheel
5. Dimensions of the Flywheel Rim

Chapter 4. C Governor

1. Define - Height of a governor, Equilibrium speed, Mean equilibrium speed & Maximum and minimum equilibrium speeds.
2. Sensitiveness of Governors
3. Stability of Governors
4. Isochronous Governors
5. Hunting
6. Effort and Power of a Governor
7. Coefficient of Insensitiveness